



GDPR Personal Data Handling

A Comprehensive Guide for
Application Developers

Prepared by JIBE



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D2.2 - The FOODITY infrastructure including the SD- and DD-Kit v2

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The FOODITY project

The main goal of FOODITY is to create a European ecosystem of digital solutions that contributes to a more sustainable and healthy food and nutrition system — respecting citizens' rights to personal data sovereignty.

FOODITY will run a **2M€ pilot development programme for creating 12 data-driven solutions** to demonstrate the potential of data-driven innovation in food and nutrition while guaranteeing the user's full control and ownership over their personal data.

These innovations will provide novel solutions to existing research and technical challenges to achieve the four Food 20301 (the EU's research and innovation policy to build better food systems by 2030) key priorities: (1) Nutrition for sustainable and healthy diets, (2) Food systems supporting a healthy planet, (3) Circularity and resource efficiency, and (4) Innovation and empowering communities.

Participants in the pilot development programme will be selected through two open calls to be launched in July and October 2023, respectively. Eligible beneficiaries will be research and technology organisations (RTOs) and universities, small and medium-sized enterprises (SMEs), social innovation actors and training organisations.

The programme will provide its beneficiaries with a set of value-added services: from one-to-one mentoring to training on technical, business, and legal matters, as well as technical and financial support (up to 187.500€ per beneficiary) for developing their pilots.

During the next 3 years (January 2023 - December 2025), the FOODITY consortium, comprised of 7 partners, will join forces to make this a reality.



Executive Summary

This report, "GDPR Personal Data Handling: A Comprehensive Guide for Application Developers," offers guidance on complying with the General Data Protection Regulation (GDPR), particularly for application developers integrating with the DataU platform. It emphasizes that personal data is broadly defined under GDPR, encompassing any information that can identify an individual, including seemingly innocuous data like food preferences and purchase history when linked to a person.

The report distinguishes between truly anonymous data (not covered by GDPR) and pseudonymized data (still considered personal data), highlighting that genuine anonymization is complex. It stresses the importance of a legal basis for initial data collection, even if the data is later anonymized.

A significant portion is dedicated to GDPR compliance throughout the application lifecycle, covering:

- **Design Phase (Privacy by Design):** Data minimization, purpose limitation, privacy controls architecture, and security by design.
- **Build Phase (Implementation Best Practices):** Granular consent management, implementation of data subject rights (access, rectification, erasure, portability, objection), robust security measures, transparency, documentation, and data retention policies.
- **Operations Phase (Ongoing Compliance):** Monitoring, auditing, breach response procedures, third-party management, training, and continuous improvement.

The report then introduces the DataU Platform as a revolutionary solution for GDPR compliance. DataU shifts from organization-controlled to user-controlled data storage and permission management through a distributed architecture featuring nodeU for permissions, proxyU for secure data transport, idU for identity verification, and dashboardU for user management.

DataU addresses specific GDPR requirements by:

- Providing blockchain-based consent management.
- Enabling instant user access and portability of data.
- Simplifying the right to erasure (organizations lose access when permissions are revoked).
- Promoting data minimization by design.
- Enhancing security through distributed storage and encrypted connections.
- Reducing breach risk and notification burdens.
- Automating records of processing activities.
- Streamlining third-party processor management.

The document highlights key technical features like blockchain-based permission management for immutability and crash/byzantine fault tolerance, alongside robust security measures. It details benefits for developers, organizations, and data subjects, including reduced compliance overhead, lower liability, increased user trust, and true data control.

Finally, the report concludes by reinforcing that DataU transforms GDPR compliance from a burden into a strategic advantage, offering significant cost savings and competitive positioning in a privacy-conscious



market. It recommends DataU integration as a foundation for future-proof digital services and provides integration steps and real-world use cases across e-commerce, e-health, e-mobility, IoT, smart cities, and government services.

Appendices include key GDPR articles, a glossary of terms, and additional resources.



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1 What is Personal Information under GDPR?

1.1 Definition

Under the General Data Protection Regulation (GDPR), personal information (or "personal data") is defined as:

Any information relating to an identified or identifiable natural person (called a "data subject").

This means information that can be used to directly or indirectly identify someone. A person is "identifiable" if they can be identified by reference to an identifier such as:

- A name
- An identification number
- Location data
- An online identifier
- One or more factors specific to their physical, physiological, genetic, mental, economic, cultural, or social identity

1.2 Key Points

The **Scope is very broad**: GDPR's definition is intentionally wide-ranging. It includes obvious identifiers but also data that could identify someone when combined with other information.

Direct and indirect identification: Even if data doesn't directly name someone, it's still personal data if it can be used to single them out.

Example: Online Bookstore

Let's say you run an online bookstore and collect the following about a customer:

- Name: Sarah Martinez
- Email: sarah.martinez@email.com
- Home address: 123 Oak Street, Manchester, UK
- Purchase history: Bought 3 cookbooks and 2 mystery novels
- IP address: 192.168.1.1
- Browser cookie ID: abc123xyz789

All of this is personal information under GDPR because it relates to an identifiable person (Sarah). Even the cookie ID and IP address alone would be considered personal data because they can be used to identify her device and potentially link back to her.

Even if you only stored "Customer in Manchester who bought cookbooks on March 15th" this could still be personal data if there's a reasonable way to identify which specific person this refers to.



2 Food Preferences and Purchase Data as Personal Information

2.1 Yes, Food Data is Personal Data!

Information about the food someone likes, buys, or eats is **personal data under GDPR**, as long as it relates to an identifiable person.

2.2 Why Food Data Matters

Food preferences and purchases reveal information about a person's:

- **Tastes and lifestyle:** Vegetarian, health-conscious, etc.
- **Cultural or religious identity:** Halal, kosher foods
- **Health conditions:** Gluten-free, diabetic-friendly foods
- **Economic status:** Budget vs. premium products
- **Household composition:** Baby food = likely has children

This falls under GDPR's definition because it's information "relating to" an identifiable person and reveals aspects of their behavior, preferences, and potentially their physical or cultural characteristics.

2.3 Examples

Clearly personal data:

- "John Smith purchased organic vegetables and lactose-free milk on Tuesday"
- "Customer ID 12345's shopping basket contains vegan products"
- A loyalty card showing Sarah's purchase history of gluten-free items

Still personal data (if linkable to a person):

- Even anonymized purchase data like "User ABC bought halal meat" - if that user ID can be connected back to a real person through other data

Special Category Consideration

If food data reveals health information (like buying diabetic foods, allergy-free products, or medical nutrition), it could even be considered "special category data" under GDPR Article 9, which has stricter protection requirements.



3 When Data Does Not Fall Under GDPR

Food preferences or purchase data would **not** fall under GDPR in these situations:

3.1 Truly Anonymous Data

If the data is genuinely anonymized so that the person cannot be identified, even indirectly:

- **Aggregate statistics:** "30% of customers buy organic vegetables"
- **Properly anonymized datasets** where re-identification is not reasonably possible

Important: Simple removal of names isn't enough; if you can still identify someone through other data points (like unique purchase patterns), it's still personal data.

3.2 Data About Legal Entities (Not Natural Persons)

- **Business purchases:** "ABC Corporation ordered 500 sandwiches for their conference"
- Company cafeteria orders from a catering business

GDPR protects natural persons (individuals), not companies or organizations.

3.3 Data of Deceased Persons

GDPR only applies to living individuals. However, some EU Member States have national laws that extend protections to deceased persons' data.

3.4 Purely Personal/Household Activity

If you're keeping track of food data for purely personal use:

- Your personal grocery shopping list on paper at home
- A private diary of what you ate

However, If you use an app or service, that company processing **your data is subject to GDPR**.

3.5 Outside EU/EEA Scope

GDPR doesn't apply if:

- The data subject is not in the EU/EEA
- The business doesn't operate in or target the EU/EEA market
- No goods/services are offered to EU residents

**Examples:**

✗ **Still GDPR-protected:** "We removed customer names but kept their purchase history linked to email addresses" > **still identifiable**

✓ **Not GDPR-protected:** "25% of all transactions included gluten-free products" > **truly anonymous aggregate data**

3.6 Anonymization vs. Pseudonymization

Even without names and emails, purchase history can still identify people if:

- **Unique purchase patterns:** Someone who bought unusual combinations can be singled out
- **Combined with other data:** Purchase timestamps + store location could identify someone
- **Rich purchase history:** The more detailed the history, the easier to identify

GDPR Classification:

- **Pseudonymized data (not anonymized):** If you removed direct identifiers but the data could still be linked back to individuals (even with effort), it's pseudonymized. This is STILL personal data under GDPR.
- **Truly anonymized data:** Only if re-identification is not reasonably possible by anyone (including you) and the process is irreversible. This is NOT personal data under GDPR.

3.7 Consent for Anonymization

Short Answer: No, you don't need consent for anonymization itself - but you DO need a legal basis for the initial processing.

Key Points:

- **Anonymized Data is NOT Personal Data:** Once data is truly anonymized (irreversibly), GDPR no longer applies to that data. You don't need consent to use anonymized data.
- **BUT - You Need Legal Basis for Initial Collection:** Before you can anonymize data, you must first collect and process the personal data, which REQUIRES a legal basis under GDPR Article 6 (consent, contract, legal obligation, legitimate interests, vital interests, or public task).
- **Transparency Requirement:** Even if you don't need specific consent for anonymization, you should inform people in your privacy policy about what you'll do with their data, including anonymization.



4 Handling Consumer Data in Application Development

4.1 GDPR Compliance Throughout the Application Lifecycle

As an application developer, you must integrate GDPR principles at every stage: design, build, and operations.

4.1.1 A. Design Phase: Privacy by Design

1. Data Minimization

- Only collect data that is necessary for your service
- Question every data field: "Do we really need this?"
- Avoid "nice to have" data collection

2. Purpose Limitation

- Define clear purposes for data collection before development
- Document why each piece of data is needed
- Don't repurpose data without new consent

3. Privacy Controls Architecture

- Design consent management interfaces early
- Plan for data subject rights (access, deletion, portability)
- Architecture should support data isolation and deletion

4. Security by Design

- Plan encryption (at rest and in transit) from the start
- Design secure authentication and authorization systems
- Consider pseudonymization and anonymization strategies

4.1.2 B. Build Phase: Implementation Best Practices

1. Consent Management Implementation

- Implement granular consent options (not just "accept all")
- Make consent freely given, specific, informed, and unambiguous
- Store consent records with timestamps and scope details
- Enable easy consent withdrawal functionality



2. Data Subject Rights Implementation

- Right to Access: API/interface for users to download their data
- Right to Rectification: Allow users to correct inaccurate data
- Right to Erasure: Implement "delete my account" functionality
- Right to Data Portability: Export data in machine-readable format (JSON, CSV)
- Right to Object: Allow users to stop specific processing activities

3. Security Implementation

- Use TLS/SSL for all data transmission
- Encrypt sensitive data at rest (AES-256 recommended)
- Implement role-based access control (RBAC)
- Use secure password hashing (bcrypt, Argon2)
- Regular security testing and vulnerability assessments

4. Transparency & Documentation

- Create clear, readable privacy policies
- Implement cookie banners with real choices
- Provide transparency about data processing activities
- Document all data flows and processing activities

5. Data Retention Implementation

- Implement automated data deletion schedules
- Don't keep data longer than necessary
- Archive or anonymize old data instead of deleting if needed for legitimate purposes

4.1.3 C. Operations Phase: Ongoing Compliance

1. Monitoring & Auditing

- Log all access to personal data for audit trails
- Regular Data Protection Impact Assessments (DPIAs)
- Monitor for unauthorized access or breaches
- Periodic compliance audits



2. Breach Response Procedures

- Have a breach response plan ready
- Must notify supervisory authority within 72 hours of breach discovery
- Notify affected users if breach poses high risk
- Document all breaches even if not reportable

3. Third-Party Management

- Vet all third-party processors for GDPR compliance
- Have Data Processing Agreements (DPAs) in place
- Ensure third parties only process data per your instructions
- Regularly review third-party compliance

4. Training & Awareness

- Regular GDPR training for all staff handling data
- Keep team updated on regulatory changes
- Foster a privacy-first culture

5. Continuous Improvement

- Regular reviews of data collection practices
- Update privacy policies as processes change
- Implement user feedback on privacy controls
- Stay current with GDPR guidance and case law



Quick Developer Compliance Checklist

- ☐ Legal basis identified for all data processing
- ☐ Data minimization principles applied
- ☐ Consent mechanisms implemented
- ☐ Privacy policy created and accessible
- ☐ Data subject rights functionality built
- ☐ Encryption implemented (transit & rest)
- ☐ Access controls and authentication secure
- ☐ Audit logging enabled
- ☐ Data retention policies automated
- ☐ Breach response plan documented
- ☐ DPAs with all processors signed
- ☐ Regular compliance reviews scheduled



5 How DataU Platform Integration Solves GDPR Challenges

5.1 DataU: Revolutionary GDPR Compliance Solution

DataU fundamentally transforms GDPR compliance by shifting from organization-controlled to **user-controlled data storage and permission management**.

5.2 The DataU Paradigm Shift

Approach	Description
Traditional Model	Organizations store user data → Users request access/deletion → Complex compliance burden
DataU Model	Users store their own data → Grant/revoke permissions to organizations → Automatic compliance

5.3 DataU Architecture Overview

DataU is a distributed platform designed to be secure, scalable, and performant. By design, there is no central data storage or processing entity. The architecture consists of several key components:

5.3.1 Core Components

1. **nodeU (DATAU Node):** The primary building block that maintains the distributed network on behalf of data subjects and data processors. Through nodeU, permissions can be registered and administered. It exposes an API for consent handling and data exchange initialization.
2. **proxyU (Data Transport):** Runs on data processor premises and sets up peer-to-peer, secure tunnels to other proxyU instances after verifying authentication and authorization through nodeU.
3. **idU (Identity Application):** Handles identity verification using a public key infrastructure (PKI) with DataU certificate authority (CA) at its root. Identity providers operate idU instances to correctly identify nodeU operators, data subjects, and data processors.
4. **dashboardU:** Reference client application that provides the user interface for data subjects to manage their permissions and data sharing.



5.3.2 How DataU Solves Specific GDPR Requirements

GDPR Requirement	Traditional Approach	DataU Solution
Lawful Basis & Consent Management	Complex consent databases, cookie banners, difficult to track consent scope and withdrawal	Built-in blockchain-based consent management. Users grant/revoke permissions through single dashboard. All consent immutably logged with timestamps.
Right to Access (Art. 15)	Manual data extraction, multiple systems to query, response delays	Users already have their data in their own DataU wallet. Instant access, no requests needed.
Right to Erasure (Art. 17)	Complex deletion across multiple databases, backup considerations, third-party coordination	User revokes permission → Organization immediately loses access. No data deletion burden on organization.
Right to Data Portability (Art. 20)	Export features to build, format standardization challenges	Data already in user-controlled wallet, easily portable between services through standard APIs.
Data Minimization (Art. 5)	Requires organizational discipline, tendency to over-collect	Organizations only access data users explicitly share. Natural minimization through permission model.
Security of Processing (Art. 32)	Organization responsible for securing large databases (attractive hacking targets)	Distributed storage means no central data honeypot. Peer-to-peer encrypted connections. PKI-based authentication.
Data Breach Notification (Art. 33-34)	Severe penalties for breaches, complex notification requirements	Reduced breach risk (no central storage). If organization breached, user data not exposed since stored in user wallet.
Data Protection by Design (Art. 25)	Requires architectural planning, often retrofitted	Privacy-first architecture built into platform. Distributed, user-controlled by default.
Records of Processing Activities (Art. 30)	Manual documentation, spreadsheets, compliance tracking overhead	Blockchain automatically records all permission grants/revocations. Complete audit trail built-in.
Third-Party Processor Management	DPA's with every vendor, monitoring compliance, liability chain	User controls which services access their data. Organizations share responsibility with users, reducing liability.



5.4 Key Technical Features

5.4.1 Permission Management via Blockchain

DataU uses blockchain technology to make permission handling truly distributed. This provides:

- **Crash fault tolerance:** Permissions are copied to every network participant, so they're never lost
- **Byzantine fault tolerance:** All handling is done in the open, making fraud easily detectable
- **Immutability:** Complete audit trail of all consent decisions
- **Privacy protection:** Only non-reversible signatures are stored on the blockchain
- **Scalability:** Shared network topology with shards for different topics/sectors

5.4.2 Security Features

- Public key identification for all data processors and subjects
- Certificate-based verification through PKI
- Peer-to-peer secure tunnels for data transport
- No central data storage vulnerability
- ISO/IEC 27001:2013 Certified

5.5 Integration Benefits

5.5.1 For Developers

- Simple API integration (RESTful)
- SDK and development kits provided (DDK and SDK)
- Reduced database complexity (no personal data storage)
- Automatic consent management
- Built-in audit trails
- Standardized access across services

5.5.2 For Organizations

- Massive reduction in compliance overhead
- Lower data breach liability
- Reduced storage and infrastructure costs
- Automatic GDPR compliance
- Increased user trust and transparency
- Simplified third-party data management



5.5.3 For Data Subjects (Users):

- One central dashboard to manage all data sharing
- Instant control - grant or revoke permissions with single click
- Complete transparency about who accesses their data
- Data stays in their control (in their wallet)
- True data portability between services
- GDPR rights exercised automatically

5.6 Cost Savings with DataU

- Data Protection Officer (DPO) costs reduced
- Compliance audit expenses minimized
- Legal consultation fees decreased
- Data breach insurance premiums lowered
- Storage costs for personal data eliminated
- Manual consent management systems unnecessary
- Data subject request processing overhead removed
- Third-party processor auditing simplified

5.7 Integration Implementation Steps

1. **Choose Integration Model:** Select from Development Platform (source code), DataU as a Service (API), or Custom Solution
2. **Set Up NodeU Instance:** Deploy nodeU to manage permissions for your organization
3. **Integrate API/SDK:** Use DataU SDK or REST API to connect your application
4. **Configure Permission Scopes:** Define what data your application needs and why
5. **Update User Onboarding:** Guide users to connect their DataU wallet/dashboard
6. **Migrate Existing Data:** Transfer existing user data to their DataU wallets (with consent)
7. **Test Permission Flows:** Verify grant/revoke mechanisms work correctly
8. **Go Live & Monitor:** Launch and monitor permission management through analytics



5.8 DataU in Practice: Real-World Use Cases

Use Case	Implementation
e-Commerce	Customers manage purchase history and preferences. Share with retailers as needed. Easy portability between stores.
e-Health & Wellbeing	Personal health data stored in user wallet. Share with doctors/apps selectively. Full medical history control.
e-Mobility/EV Charging	Users control charging history, location data. Share with providers for billing. Switch providers seamlessly.
IoT Devices	Device data flows to user wallet first. Users grant access to manufacturers/services. Full control over sensor data.
Smart Cities	Citizens control data shared with government services. Transparent, trust-building civic engagement.
Loyalty Programs	Points and preferences in user wallet. Use across multiple retailers. True data portability.
Government Services (H2020 PoSeID-on)	GDPR-compliance application designed for millions of private users sharing data with central services of governmental organizations.

5.9 DataU Service Application Store

DataU offers a range of pre-built applications that developers can leverage:

- **e-IOT:** Easily connects IoT devices and manages data-sharing with external services
- **e-Commerce:** Manages data-sharing between customers and e-commerce applications
- **e-Mobility:** Facilitates data-sharing between stakeholders in the EV-charging value chain
- **e-Health & e-Wellbeing:** Dedicated application for monitoring personal health and wellbeing
- **e-Energy:** Controls and optimizes energy consumption in private homes to save costs

Three Integration Routes

1. **DataU as a Development Platform:** DataU source code is available for ICT companies/departments or research groups that wish to develop further services on top.
2. **DataU as a Service:** DataU as a data sharing & permission network service, to connect your own e-services. Easy Plug and Play instructions, API and examples supplied to developers.
3. **DataU Customer-Specific Solutions:** Custom developmental projects to design & build customer-specific solutions.



6 Conclusion

GDPR compliance represents a significant challenge for application developers and organizations. The regulation's broad scope, strict requirements, and substantial penalties (up to €20 million or 4% of global annual turnover, whichever is higher) demand careful attention throughout the entire application lifecycle.

6.1 Key Takeaways from This Report:

1. **Personal Data is Broadly Defined:** GDPR's definition encompasses not just obvious identifiers like names and addresses, but also behavioural data, preferences, and even food choices when linked to identifiable individuals.
2. **Anonymization Must Be Genuine:** Simply removing names is insufficient. True anonymization requires making re-identification practically impossible, while pseudonymization still falls under GDPR protection.
3. **Compliance Requires Lifecycle Integration:** GDPR principles must be embedded from the design phase through build and operations. Privacy by design is not optional—it's required by Article 25.
4. **Data Subject Rights Are Mandatory:** Applications must provide mechanisms for users to access, rectify, erase, port, and object to processing of their personal data.
5. **Traditional Compliance is Complex and Costly:** Organizations face significant ongoing expenses for DPOs, legal consultations, audits, security infrastructure, and breach management.

6.2 The DataU Advantage

However, **DataU offers a revolutionary solution** that transforms GDPR compliance from a burden into a competitive advantage. By shifting to a user-controlled data architecture:

- **Developers** can build applications without the complexity of managing personal data storage and compliance infrastructure
- **Organizations** dramatically reduce their compliance costs, breach liability, and operational overhead while increasing user trust
- **Users** gain true control over their personal data with transparency and easy management through a single dashboard

6.3 Why DataU Works

The DataU platform's distributed architecture, blockchain-based permission management, and user-centric design address the fundamental challenge of GDPR compliance: **the conflict between organizational data needs and individual privacy rights.**

By inverting the traditional model—where organizations control data and users request access—DataU creates a system where:



- Users naturally control their own data (stored in their wallet)
- Organizations access only what users explicitly permit
- Consent is automatically managed via blockchain
- Data subject rights are exercised instantly without organizational burden
- Compliance becomes an architectural feature rather than an operational challenge

6.4 Practical Impact

Metric	Traditional Approach	With DataU Solution
Time to process access request	Up to 30 days (legally required response time)	Instant (user already has data)
Cost of data breach	€4.4M average + penalties up to €20M	Significantly reduced (no central storage)
Consent management complexity	Custom systems, legal review, tracking infrastructure	Built-in, blockchain-verified, automatic
Data deletion complexity	Multiple systems, backups, third-party coordination	Permission revocation = instant access removal
Annual compliance costs	€100K-€1M+ depending on organization size	70-90% reduction in ongoing costs

6.5 Industry Applications

DataU has proven applications across multiple sectors:

- **B2B:** Industrial value chains with cross-border data transfers for Industry 4.0 solutions
- **B2C:** E-commerce, health, mobility, and energy services with direct consumer data management
- **B2G2C:** Government-citizen services including smart cities and public administration platforms

Real implementations include GDPR-compliance applications for millions of users in H2020 projects, data sharing networks for industrial supply chains, and loyalty point systems across multiple retailers.

6.6 Strategic Recommendation

For organizations building new applications, DataU integration should be seriously considered as a strategic foundation for privacy-first, regulation-compliant digital services. The platform offers:

- Reduced time-to-market (no need to build complex compliance infrastructure)
- Lower total cost of ownership (reduced infrastructure, legal, and operational costs)
- Enhanced competitive positioning (user trust through genuine data control)
- Future-proof architecture (ready for evolving privacy regulations)



- Scalability without proportional compliance cost increases

6.7 Looking Forward

As privacy regulations continue to evolve globally—with frameworks like California's CCPA, Brazil's LGPD, and others following GDPR's lead—the traditional model of centralized data control becomes increasingly untenable.

DataU's user-controlled, distributed architecture represents not just a GDPR compliance solution, but a fundamental reimagining of how digital services can operate in a privacy-conscious world. Organizations that adopt this model early will benefit from:

- Competitive advantage in privacy-conscious markets
- Reduced regulatory risk as laws evolve
- Lower operational costs over time
- Stronger user relationships built on trust and transparency
- Easier expansion into new jurisdictions with varying privacy laws

6.8 Final Thoughts

GDPR compliance need not be viewed solely as a regulatory burden. With the right architectural approach—exemplified by DataU—it becomes an opportunity to build better, more trustworthy digital services that respect user privacy while enabling innovation.

The question for organizations is no longer whether to comply with GDPR, but **how** to comply most effectively. DataU provides a clear answer: by giving users genuine control over their data, organizations can achieve compliance automatically while focusing on delivering value through their core services.

6.9 Next Steps for Implementation

Organizations interested in DataU integration should:

- Assess current data handling practices and GDPR compliance gaps
- Evaluate which DataU integration model best fits organizational needs
- Conduct a pilot project with a specific use case
- Measure cost savings and compliance improvements
- Plan phased rollout across services and applications

Contact: For more information about DataU integration, visit www.jibecompany.com or contact the DataU team.



Appendix A: GDPR Key Articles Reference

Article	Key Provisions
Article 4	Definitions including personal data, processing, controller, processor, consent
Article 5	Principles: lawfulness, fairness, transparency, purpose limitation, data minimization, accuracy, storage limitation, integrity, confidentiality, accountability
Article 6	Lawful basis for processing: consent, contract, legal obligation, vital interests, public task, legitimate interests
Article 9	Special categories of personal data (health, genetic, biometric, racial, political, religious, etc.)
Article 15	Right of access by the data subject
Article 16	Right to rectification
Article 17	Right to erasure ("right to be forgotten")
Article 18	Right to restriction of processing
Article 20	Right to data portability
Article 21	Right to object
Article 25	Data protection by design and by default
Article 30	Records of processing activities
Article 32	Security of processing
Article 33	Notification of personal data breach to supervisory authority (72 hours)
Article 34	Communication of personal data breach to data subject
Article 35	Data protection impact assessment (DPIA)
Article 83	Administrative fines (up to €20M or 4% of global annual turnover)



Appendix B: Glossary of Terms

Term	Definition
Anonymization	Irreversible process rendering data non-identifiable; result is no longer personal data
Consent	Freely given, specific, informed, and unambiguous indication of agreement to processing
dashboardU	DataU user interface for data subjects to manage permissions and data sharing
Data Controller	Entity that determines the purposes and means of processing personal data
Data Processor (DP)	Entity that processes personal data on behalf of the controller
Data Subject (DS)	An identified or identifiable natural person whose personal data is processed
DPA	Data Processing Agreement - contract between controller and processor
DPIA	Data Protection Impact Assessment - required for high-risk processing
DPO	Data Protection Officer - required for certain organizations under GDPR
idU	DataU identity verification application using PKI certificates
nodeU	DataU network node maintaining distributed permissions between data subjects and processors
Personal Data	Any information relating to an identified or identifiable natural person
Processing	Any operation performed on personal data (collection, storage, use, disclosure, deletion, etc.)
proxyU	DataU application for peer-to-peer secure data transport between processors
Pseudonymization	Processing to make data non-attributable without additional information; still personal data
Supervisory Authority	Independent public authority monitoring GDPR compliance in each EU member state



Appendix C: Additional Resources

Official GDPR Resources

- **Full GDPR Text:** <https://gdpr-info.eu/>
- **European Data Protection Board:** <https://edpb.europa.eu/>
- **EU Commission GDPR Page:** https://ec.europa.eu/info/law/law-topic/data-protection_en

DataU Resources

- **Jibe Company (DataU Developer):** www.jibecompany.com
- **DataU Platform Information:** Available through Jibe Company contact
- **Contact:** Hans Houf, Executive Advisory Board - hans.houf@jibecompany.com
- **Phone:** +359 898 455 955

GDPR Guidance Documents

- EDPB Guidelines on Consent
- EDPB Guidelines on Data Protection by Design and by Default
- EDPB Guidelines on Data Protection Impact Assessment
- EDPB Guidelines on Personal Data Breach Notification
- ICO (UK) Guidance on various GDPR topics

Technical Standards

- **ISO/IEC 27001:2013** - Information Security Management (DataU is certified)
- **ISO/IEC 27701:2019** - Privacy Information Management
- **ISO/IEC 29100:2011** - Privacy Framework